

Engineering Technical Log: Candidate Generation & Blocking Pipeline

Role: Member 3 (Candidate Generation & Blocking) | **Challenge:** Amazon ML Challenge (Entity Resolution)
Environment: macOS Apple Silicon (M4), DuckDB 1.x, Python 3.9 | **Deliverable:** output/candidate_pairs.tsv

1. Problem Definition & Challenge Scope

The unlabeled Test dataset contains **1,732,544** Source 1 records against ~10,000,000+ Target pool records (Source 2 + Source 3), creating an unconstrained search space of ~**17.3 Trillion pairwise comparisons**. Member 3 engineered an out-of-core blocking pipeline to maximize candidate recall while strictly capping candidates to ≤15 per entity.

2. Chronological Pipeline Engineering Log

Step 1: Zero Cross-Country Constraint

Training ground truth confirmed 0.00% cross-border matches. Hard geographic partitioning (France, United States, India) was established as the primary search isolation boundary.

Step 2: 8-Route Inverted Blocking Index Design

- **Route 1 (Exact Normalized Name):** Exact match on cleaned business names (>=3 chars).
- **Route 2 (Distinctive Token 1):** Matches first token stripped of business stopwords; frequency capped at 800.
- **Route 3 (Cross-Token 1 <-> 2):** Matches S1 Token 1 with Target Pool Token 2 (>=4 chars).
- **Route 4 (Cross-Token 2 <-> 1):** Matches S1 Token 2 with Target Pool Token 1 (>=4 chars).
- **Route 5 (Distinctive Token 2):** Matches second word, handling variations in leading qualifiers.
- **Route 6 (Distinctive Token 3):** Matches third token for extended corporate entity names.
- **Route 7 (Address Prefix Match):** Matches leading 9 alphanumeric characters of clean addresses.
- **Route 8 (Street Anchor + Stem):** Matches exact street numbers paired with a 3-character name stem.

Step 3: Ground Truth Benchmarking

Validated against a 50,000 ground-truth sample, achieving a verified **90.69% Candidate Recall Ceiling**.

Step 4: Memory Optimization & Out-of-Core Execution

- **Spill Resolution:** Dropped the redundant 4-character prefix route that caused 208 GB disk spills.
- **Sort OOM Resolution:** Replaced monolithic window deduplication with **250k-chunk batching** and dynamic inline capping (skipping entities with 15 matches), keeping peak RAM strictly **under 2.5 GB**.

3. Production Results & Validation Audits

Metric / Dimension	Target	Achieved Production Value
Search Space Reduction Ratio	> 99.9%	99.99986% (17.3T -> 24.64M pairs)
Candidate Recall Ceiling	>= 85.0%	90.69% (Verified on Ground Truth)
Total S1 Entities Processed	1,732,544	1,732,544 (100.00%)
Entities with Valid Candidates	> 98.0%	1,726,332 (99.64%)
Total Candidate Pairs Generated	N/A	24,643,067 pairs
Candidate Density (per entity)	<= 15 max	14.23 average (Strictly <= 15 cap)
Peak Working Memory	< 16 GB	< 2.5 GB RAM (Zero spill)
End-to-End Execution Time	< 60 mins	33.58 minutes (2,015.09s)

Validation Checklist Passed:

- ✓ **Row Count Integrity:** Exactly 1,732,544 rows matching test entities.
- ✓ **Zero Empty Candidate Rows:** 0 empty rows (0.00% missing).
- ✓ **Strict Capping Enforcement:** Max candidates per row is 15 (No truncation exceptions).
- ✓ **Syntax Standard:** Tab-delimited TSV matching header `source1_entity_id\tcandidate_entity_ids`.

4. Downstream Handoff to Member 4

The generated deliverable output/candidate_pairs.tsv is fully verified and packaged. Member 4 can immediately consume this file as the candidate target space to execute pair-level feature extraction (Jaro-Winkler, Levenshtein distance, embedding cosine similarities) and classification modeling.